GWEN AI ASSISTANT

Autonomous Windows Desktop & Cloud Integration System

Portfolio Showcase
Developer: Yared Uquillas
Target: UpWork Portfolio

GWEN AI is an autonomous, high-performance virtual assistant designed for comprehensive management and automation of Windows 11 desktop environments. The project features a multi-interface architecture (Telegram Bot, FastAPI REST/WebSocket API server, and a native Android companion app in Kotlin), enabling remote control, real-time system telemetry, neural voice processing, and secure execution of automated tasks.

CORE SYSTEM CAPABILITIES & MODULES

Module / Capability	Technical Description	Commands / Interface
■ **Hardware & Power Telemetry**	Real-time remote monitoring of CPU, RAM, disk usage (C:\), battery charge state, and system uptime.	`/bateria , /sistema`
■ **Master Volume & Media Control**	Remote management of Windows audio volume and media playback control (Play, Pause, Next, Previous).	`/volumen , /play , /pausa , /next , /prev`
■ **Screen Capture & Native OCR**	Instant desktop screenshot capture with on-screen text extraction powered by native Windows OCR engines.	`/captura`
■■ **File & Cloud Management**	Downloads folder organization, automated large file uploads to Google Drive and temporary cloud hosts.	`/descargas , /drive , /transfer`
■■ **Cloud Gaming Automation**	Instant game launching and queue state tracking on NVIDIA GeForce NOW using a zero-CPU state machine.	`/lanza`
■ **Secure Command Shell**	Isolated PowerShell/CMD command execution with metacharacter sanitization to mitigate Remote Code Execution (RCE) risks.	`/sh`
■ **Autonomous Task Scheduler**	Background scheduled task management with automated execution and status notifications.	`/tareas`
■■ **Neural Voice & Speech STT/TTS**	Telegram voice note transcription and neural text-to-speech response synthesis using Microsoft Edge TTS.	`Voice In / Neural Audio Out`

TECHNICAL ARCHITECTURE & TECHNOLOGY STACK

Component	Technologies Used	Architectural Highlights
Core Backend	Python 3.11, PyTelegramBotAPI, FastAPI, Uvicorn	Event-driven modular plugin architecture with interface decoupling.
AI Engines	OpenRouter API, DeepSeek R1, Groq Llama 3.3 70B	Dynamic model fallback routing with complex reasoning support.
Windows Integration	PowerShell, PyAutoGUI, PyGetWindow, PyCaw, Psutil	Win32 API integration and low-level OS subsystem control.
Database	SQLite3 (WAL Mode), JSON Repositories	Lightweight thread-safe persistence for multi-reader workloads.